CS348 Computer Networks Quiz-1, Fri 29 Aug 2025, 08.35am-09.20am, Max. Marks: 10

Name: SAMPLE ANSWERS Roll number: ____________ House: ________________

General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only.
- All answers have to be (briefly) explained. State any necessary assumptions.

Prof Mitra Nischay (Mitran) wants to design a new wireless LAN technology, called MiNi (pronounced My Ny), as a competitor to WiFi. Your job is to help him answer the following questions.

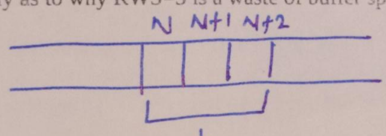
1. **[1 mark]** Give one reason why MiNi should use FEC (Forward Error Correction) and not ARQ (Automatic Repeat reQuest).

High wireless error rate

2. **[1 mark]** Give one reason why MiNi should use ARQ and not FEC.

Low propagation delay compared to transmission delay.

3. **[1 mark]** Prof Mitran has designed a link layer reliable transfer protocol with SWS=2 and RWS=3. Explain briefly as to why RWS=3 is a waste of buffer space. Draw a diagram as necessary.

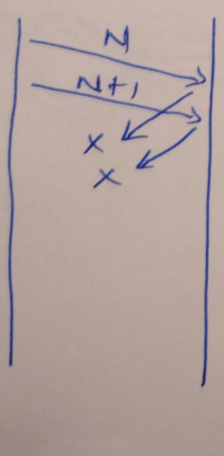


Receiver seq # space

Receiver window - upto N-1 acknowledged

N not ACKed => sender window ~~at~~ can at most be {N, N+1} => sender will not send N+2 => waste of buffer

4. **[1 mark]** Prof Mitran designed a sequence number space of size 3 (i.e. 0, 1, 2 repeating). Explain through a suitable timing diagram as to why this will lead to incorrect behaviour. Timing diagram is mandatory.



Scenario: ACKs of N & N+1 dropped

Receiver window {N+2, N+3} (not counting N+4)

When sender sends N, (retx) it will be confused with N+3 as seq. num space is only {0,1,2}

=> protocol incorrect

5. **[2 marks]** Prof Mitran has a brilliant student Sahasrabuddhi who has designed a PHY layer with capacity of 16 Mbps. But instead of using it in a LAN, Prof Mitran struck a deal with ISRO to use it to communicate with a LEO satellite. Suppose the above reliable transfer protocol is used with a frame size of 200 bytes, and the satellite is at a distance of 150km from ISRO's base on earth. What is the best case efficiency of this reliable transfer protocol? Ignore header overheads and ACK size. Show your calculation briefly.

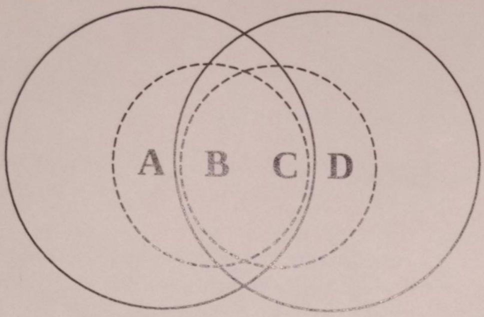
Tx Time = $(200 \times 8) / 16M = 100\ \mu s$

2x Prop delay = $300\,km / 3 \times 10^8\,m/s = 1000\ \mu s$

SWS = 2

$$\text{Efficiency} = \frac{2 \times \text{Tx Time}}{\text{Tx Time} + 2 \times PD} = \frac{200}{1100} = \frac{2}{11} \approx 18.18\%$$

6. **[1 mark]** Sahasrabuddhi's friend Ekabuddhi test deployed 4 nodes of MiNi in his office building. The diagram below shows the transmission ranges of the 4 nodes. The solid circles represent the transmission range of nodes A and D, and the dashed circles represent the transmission range of B and C.



The absent minded Prof Mitran forgot to design a MAC protocol for MiNi. Suppose A is sending data to B, while C is sending data to D. Assume both are sending packets continuously at a rate equal to the link capacity, and without any ACK mechanism. What is the received throughput at B as a percentage of A's sending rate? Explain briefly.

B is within tx range of C ⇒ all of A's pkts collide at B

Ans: 0%.

7. **[1 mark]** In the above scenario, what is the received throughput at D as a percentage of C's sending rate? Explain briefly.

D is outside tx range of A ⇒ all pkts of C successful

Ans: 100%.

8. **[2 marks]** Prof Mitran had another clever and loyal student Kotibuddhi, who said that Prof Mitran's absent mindedness (no MAC in MiNi) is a blessing in disguise. She described a scenario of two nodes transmitting in the above network (again without any ACK mechanism), while achieving twice the overall throughput compared to the case when CSMA/CA is used. Can you match Kotibuddhi's brains? Explain briefly.

B → A and C → D

With CSMA/CA they'll share 50% of link since B & C are in range of each other

⇒ total throughput = 100% link capacity

Without MAC, both B → A & C → D operate at link capacity ⇒ total throughput = 200% link capacity